الجامعة السعودية الإلكترونية
SAUDI ELECTRONIC UNIVERSITY
2011-1432

*College of Computing and Informatics*

**Computer Networks**

**DS360**

# Project-Phase 1

## Deadline: Day 26/07/2026 @ 23:59

***[Total Mark is 5]*** CRN:**50584**


Name: **Layan Alghamdi** ID: **S240014014**
Name: **Alanoud Alfalah** ID: **S240061825**
Name: **Rawan Almalki** ID: **S240051772**
Name: **Safanah Alfehaid** ID: **S240031066**


**Instructions:**

- You must submit two separate copies **(one Word file and one PDF file)** using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g., misspell words, remove spaces between words, hide characters, use different character sets, convert text into image or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You MUST show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

# Project Instruction

**Project Objective:**

Design, configure, and test a network setup in **Packet Tracer** to successfully transmit data packets between devices in different LANs across two separate WANs (WAN 1 and WAN 2).

**Group Details**

- Group Size = Maximum of 3 Members.
- One group member (group leader/coordinator) must **submit the project report** on blackboard. Marks will be given based on your submission and quality of the contents.

**Project Report**

- The project is divided into two phases:
  - Phase 1 (Mid project Progress Report)
  - Phase 2 (Final project Report)
- This report is for Phase 1. In this report, you should complete all the requirements for Phase 1 only.
- Each Report will be evaluated according to the marking criteria

*Learning Outcome(s):*

*Demonstrate protocol configuration, network-addressing schemes and analyze packet transmission for the Local and Wide Area Networks.*

## Project Description (Common to Phase 1 and Phase 2)

A university operates two campuses: a Main Campus and a Branch Campus. Each campus network is treated as a separate WAN site. Your task is to design and configure, in Cisco Packet Tracer, a network that successfully transmits data packets from a device in any LAN of WAN 1 (Main Campus) to a device in any LAN of WAN 2 (Branch Campus). Each WAN contains 2 LANs, and each LAN has 3 end devices.

---

**Network Design Requirements:**

1. **WAN 1 (Main Campus):**
   - Contains LAN 1 (Computer Lab) and LAN 2 (Admin Office).
   - Each LAN has 3 devices (PCs or laptops) connected through a switch.
   - Assign private IP addresses to the devices in each LAN (e.g., use 172.16.10.0/24 for LAN 1 and 172.16.20.0/24 for LAN 2).
   - Connect each LAN to a router (Router 1).

2. **WAN 2 (Branch Campus)**:
   - Contains LAN 3 (Library) and LAN 4 (Faculty Office).
   - Each LAN has 3 devices (PCs or laptops) connected through a switch.
   - Assign private IP addresses to the devices in each LAN (e.g., use 172.16.30.0/24 for LAN 3 and 172.16.40.0/24 for LAN 4).
   - Connect each LAN to a router (Router 2).

3. **WAN Connection**:
   - Connect Router 1 and Router 2 using a serial link to simulate a WAN connection.
   - Assign IP addresses to the serial interfaces (e.g., 10.10.10.1/30 for Router 1 and 10.10.10.2/30 for Router 2).

---

**Configuration Details:**

1. **Assign IP Addresses**:
   - Configure IP addresses, subnet masks, and default gateways for all devices in each LAN. Example:
     - LAN 1: 172.16.10.0/24, Default Gateway: 172.16.10.1 (Router 1's interface).

        - LAN 2: 172.16.20.0/24, Default Gateway: 172.16.20.1 (Router 1's interface).
        - LAN 3: 172.16.30.0/24, Default Gateway: 172.16.30.1 (Router 2's interface).
        - LAN 4: 172.16.40.0/24, Default Gateway: 172.16.40.1 (Router 2's interface).

2. **Router Configuration**:
    - Configure IP addresses on the LAN and WAN interfaces of both routers. Example:
        - Router 1:
            - LAN 1 Interface: 172.16.10.1/24
            - LAN 2 Interface: 172.16.20.1/24
            - WAN Interface (Serial): 10.10.10.1/30
        - Router 2:
            - LAN 3 Interface: 172.16.30.1/24
            - LAN 4 Interface: 172.16.40.1/24
            - WAN Interface (Serial): 10.10.10.2/30
3. **Enable Routing**:
    - Configure static routes or a dynamic routing protocol (e.g., RIP) on both routers to ensure communication between WAN 1 and WAN 2.

5 Marks

## Phase 1 Requirements

Students are required to submit a short mid progress report, which must include the following:

1. **Work completed this period (2 marks):** This includes the installation of Cisco Packet Tracer, and a study of Packet Tracer for creating nodes, using different types of links (copper straight-through, copper cross-over, serial DCE/DTE), and assigning IP addresses.

    **Note:** You must show a screenshot of the installation of Packet Tracer (with your system visible, e.g., date/time or username). (1 mark)

    Give a description of the objectives and advantages of using Packet Tracer for this project. (1 mark)
2. **One key decision made during this period (1 mark):** For example, your choice of IP addressing scheme, choice of routing method to study, or division of work among group members. Explain briefly why the decision was made.
3. **One problem encountered (1 mark):** Describe one genuine problem faced so far (technical or organizational) and its current status (resolved / in progress).
4. **Plan for the next period (0.5 mark):** Briefly list the tasks you plan to complete before the final submission (e.g., building the topology, configuring routers, testing with ping).
5. **Each group member's input/participation (0.5 mark):** State each member's name and their specific contribution during this period.

## Version History / Commit Log Requirements (Common to Both Reports)

Students must document and submit their project history/timestamps using a live, editable link:

- **Written report:** Submit a shared Google Doc or Word Online link (sharing set to "Anyone with the link can view") along with your final PDF report. The link must show the progress of the students' work with timestamps/screenshots.
- **Code/data/configuration files:** Submit Packet Tracer files (.pkt) and any configuration scripts via a GitHub or GitLab repository link (the coordinator may choose a repository suitable for the course). The repository must show a commit history reflecting incremental progress. Commit regularly (at minimum once per week), with clear, descriptive commit messages.

A separate section in the report must be added and named "Project Artifacts" to include all links, for example:

Live report document (with edit history): [link]
GitHub/GitLab repository (with commit history): [link]
Packet Tracer project file (.pkt): [link or attached]

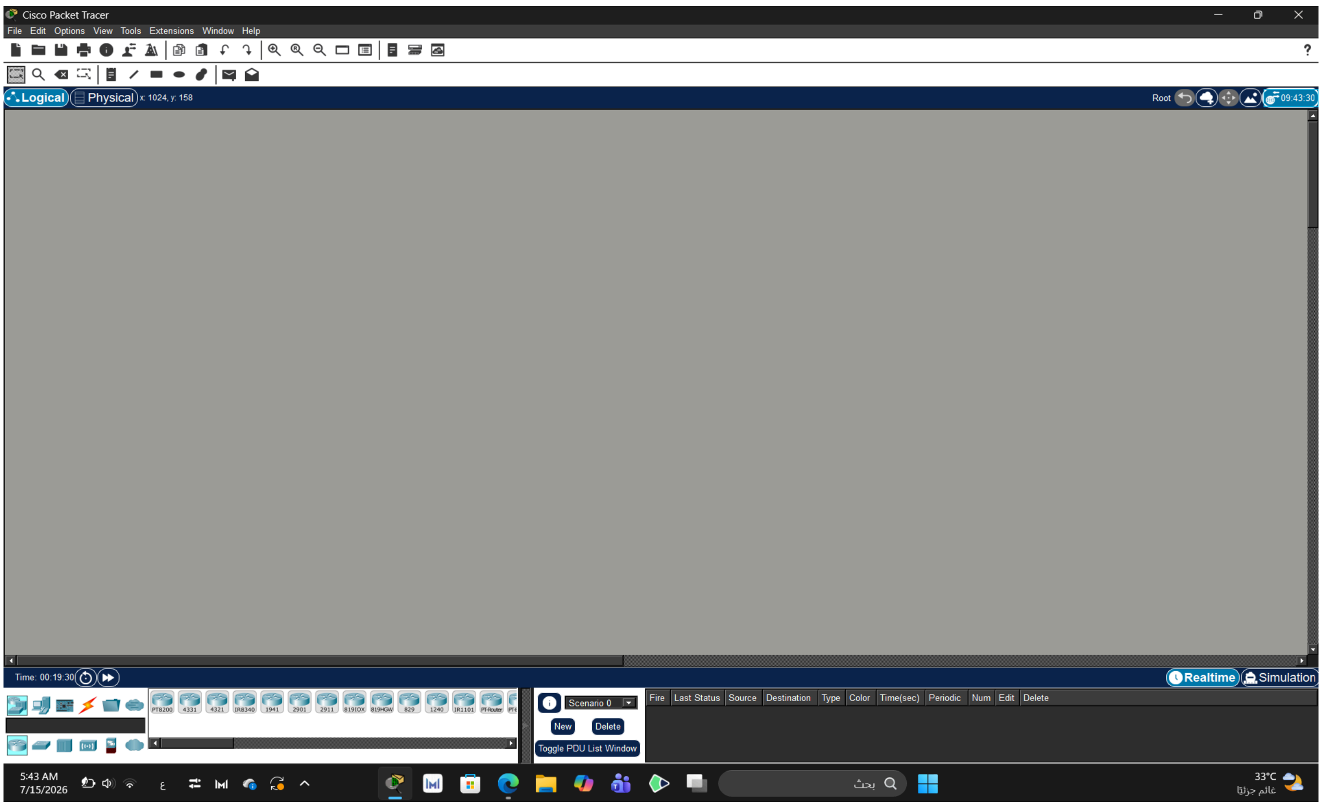
**What NOT to do:** Students must not write their report elsewhere and paste the final version in at the last minute. They also must not make a single commit containing the entire project. Both are treated as red flags regardless of the quality of the final work.

## Important Notes

- **AI use:** Students may use AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) only for brainstorming approaches, debugging small errors, or proofreading text. AI must not be used to generate analysis, write configuration logic, or produce substantial portions of the report or project files. Any permitted AI use must be disclosed in the Reflection section. Undisclosed AI use, or AI use beyond this scope (disclosed or not), will result in a zero grade for the entire project.
- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit the mid progress report as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.
- The mid progress report should be concise including screenshots.
- The same shared document link and repository submitted in Phase 1 must continue to be used for Phase 2, so that the full project history is visible in one place.

**Cisco Packet Tracer Installation**

**Screenshot** of Cisco Packet Tracer Installation



**Objectives of Using Cisco Packet Tracer:**

- Using a virtual environment to understand networking concepts and try creating different designs virtually.
- Understanding how network devices work, how to configure them, and how to connect them with each other.
- Applying and analyzing network performance and practicing solving problems before implementing them on the required devices.
- Gaining experience and developing skills in designing networks suitable for the project requirements.

**Advantages of Using Cisco Packet Tracer in the Project:**

- It provides an educational tool to experiment with network configurations without the need for real equipment.
- It makes the process of designing and modifying network layouts easier, and allows us to test changes on them.
- It helps in understanding the communication method between devices and how they interact with each other.
- It saves time and effort because it reduces errors before implementing the network design.

### One Key Decision Made

- **Decision:** We chose to use the IP addressing scheme provided in the project instructions (172.16.10.0/24, 172.16.20.0/24, 172.16.30.0/24, and 172.16.40.0/24).

- **Reason:** This makes it easy to set up the network, as each LAN has its own subnet. It helps avoid addressing errors and makes router configuration and troubleshooting easier later.

### A Problem Encountered

- **Problem:** At first, we didn't know which type of cable should be used for each connection in Packet Tracer, especially the Serial DCE/DTE cable between the two routers.

- **Status:** Resolved. We read the Packet Tracer documentation and the course materials to figure out which cable to use for each connection and continue with the project.

### Plan for the Next Period

1. Prototype the network topology on paper.
2. Build the network topology in Cisco Packer Tracer.
3. Use the correct cables to connect the devices.
4. Assign IP addresses and configure the routers.
5. Test the network in Cisco Packer Tracer.
6. Configure routing between the two WANs.
7. Test network connectivity using the ping command.
8. Troubleshoot and fix any connectivity issues before the final submission.

**Each Group Member Input/Participation**

| Student | Assigned Tasks | Expected Deliverables |
|---|---|---|
| **Layan Alghamdi** | Install **Cisco Packet Tracer**. Take the required screenshot showing the date or username. Write the objectives of using Packet Tracer. Explain the advantages of using it in the project. | Packet Tracer installation screenshot. A short paragraph about the objectives. A short paragraph about the advantages. |
| **Alanoud Alfalah** | Review the project requirements. Write **One Key Decision Made**, such as selecting the IP addressing plan or the method of dividing the work, and explain the reason for the decision. Write **One Problem Encountered** and clarify whether it has been solved or is still in progress. | Description of the key decision and its justification. Description of the problem encountered. Current status of the problem and the solution used or proposed. |
| **Rawan Almalki** | Write the **Plan for the Next Period**. Create the initial Packet Tracer file in **.pkt** format. Build the basic network topology, including Router 1, Router 2, switches, end devices, and cable connections. | A clear plan for the next project period. Packet Tracer file in **.pkt** format. An initial network topology with all required devices and connections. |
| **Safanah Alfehaid** | Prepare the **Each Group Member's Input/Participation** table and describe each student's contribution. Create a shared file using Google Docs or Word Online and share it with the group. Create a GitHub or GitLab repository and add the first commit. | Completed participation table. Link to the shared online document. Repository link and evidence of the first commit. |

## Project Artifacts

Google Doc: IT351_Project-Phase 1

GitHub/GitLab repository: GitHub

Packet Tracer project file (.pkt): Network Initial Design

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*College of Computing and Informatics*

**Computer Networks**

**IT351**

# Project-Phase 2

## Deadline: Day 09/08/2026 @ 23:59

***[Total Mark is 15]***


**Name:** Layan Alghamdi **ID:** S240014014
**Name:** Alanoud Alfalah **ID:** S240061825
**Name:** Rawan Almalki **ID:** S240051772
**Name:** Safanah Alfehaid **ID:** S240031066


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**Project Report**

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*Learning Outcome(s):*

*Demonstrate protocol configuration, network-addressing schemes and analyze packet transmission for the Local and Wide Area Networks.*

## Project Description (Common to Phase 1 and Phase 2)

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   - Connect each LAN to a router (Router 1).

2. **WAN 2 (Branch Campus)**:
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   - Each LAN has 3 devices (PCs or laptops) connected through a switch.
   - Assign private IP addresses to the devices in each LAN (e.g., use 172.16.30.0/24 for LAN 3 and 172.16.40.0/24 for LAN 4).
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3. **WAN Connection**:
   - Connect Router 1 and Router 2 using a serial link to simulate a WAN connection.
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    - Configure IP addresses on the LAN and WAN interfaces of both routers. Example:
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        - Router 2:
            - LAN 3 Interface: 172.16.30.1/24
            - LAN 4 Interface: 172.16.40.1/24
            - WAN Interface (Serial): 10.10.10.2/30
3. **Enable Routing**:
    - Configure static routes or a dynamic routing protocol (e.g., RIP) on both routers to ensure communication between WAN 1 and WAN 2.

15 Marks

## Phase 2 Requirements

### Testing and Verification

1. Use the ping command to test connectivity between a device in any LAN of WAN 1 and a device in any LAN of WAN 2.
2. Verify that data packets are successfully transmitted (0% packet loss).

### Questions to Answer (10 Marks — 2 marks each)

Answer all five questions. For each question, you must include (a) a clear screenshot from your Packet Tracer project and (b) the commands used at that instance (e.g., IOS CLI commands or the configuration steps applied). Screenshots without the corresponding commands/steps will receive ZERO marks for that question.

1. **Creating nodes:** Show a screenshot of the complete topology with all nodes created (12 end devices, 4 switches, 2 routers). List the device types selected and any steps/commands used to name the devices (e.g., hostname configuration). (2 marks)
2. **Giving connections in the LAN:** Show a screenshot of the LAN connections between end devices, switches, and routers. State the cable types used for each connection and why. (2 marks)
3. **Assigning IP addresses:** Show screenshots of IP address, subnet mask, and default gateway assignment on at least one end device in each LAN, and the commands used to configure the router LAN interfaces (e.g., interface, ip address, no shutdown). (2 marks)
4. **Creating the WAN connection:** Show a screenshot of the serial link between Router 1 and Router 2, and the commands used to configure the serial interfaces and enable routing (static routes or a dynamic routing protocol such as RIP), including clock rate if applicable. (2 marks)
5. **Successful data transmission using the ping command:** Show screenshots of successful ping tests between a device in WAN 1 and a device in WAN 2 (both directions), with the exact ping commands used and the results (replies received, 0% loss). (2 marks)

### Written Reflection — End of Project Report (5 Marks)

The project report must include a Reflection section (300–500 words) as the last section before the Project Artifacts section. This section must be written in the students' own words and should address some of the following:

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method (e.g., static routing vs. RIP, your IP addressing scheme) over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

## Version History / Commit Log Requirements (Common to Both Reports)

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- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit the mid progress report as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.
- The final report must be submitted as a PDF, together with the live document link, the repository link, and the Packet Tracer (.pkt) file.
- The edit history of the live document and the commit history of the repository must show incremental progress across both phases (Phase 1 and Phase 2).

## Network Topology

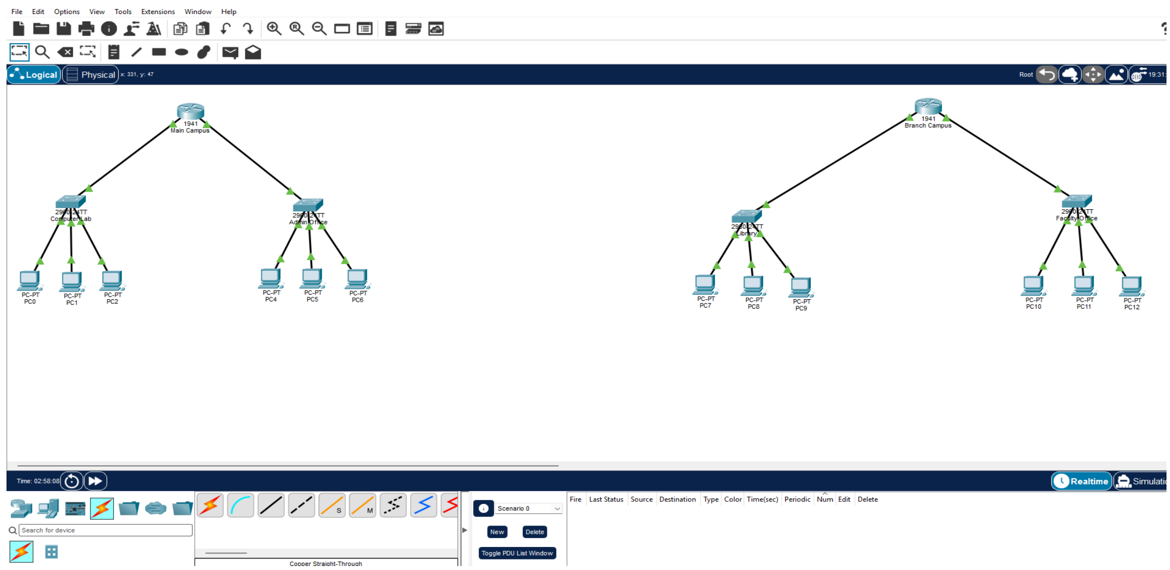



Figure 1. Initial Network Topology

## Devices Used in the Network

| Device Type | Model | Quantity |
|---|---|---|
| Router | Cisco 1941 | 2 |
| Switch | Cisco 2960-24TT | 4 |
| PC | PC-PT | 12 |

**Commands Used to Change Device Names**

Router (**Main Campus**):

enable

configure terminal

hostname MainCampus

end

copy running-config startup-config

Router (**Branch Campus**):

enable

configure terminal

hostname Branch Campus

end

copy running-config startup-config

Switch (**Computer Lab**) :

enable

configure terminal

hostname ComputerLab

end

copy running-config startup-config

Switch (**Admin Office**):

enable

configure terminal

hostname AdminOffice

end

copy running-config startup-config

Switch (**Library**):

enable

configure terminal

hostname Library

end

copy running-config startup-config

Switch (**Faculty Office**):

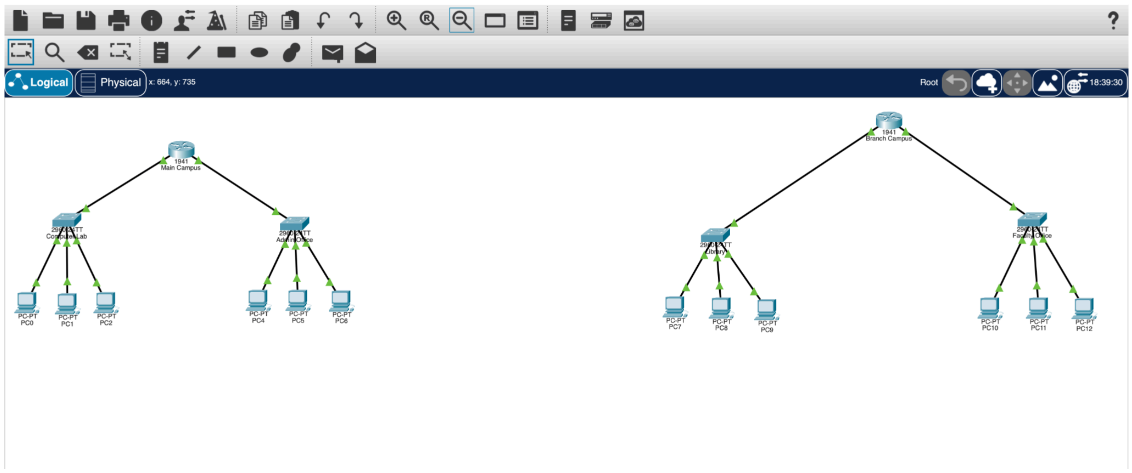
enable

configure terminal

hostname FacultyOffice

end

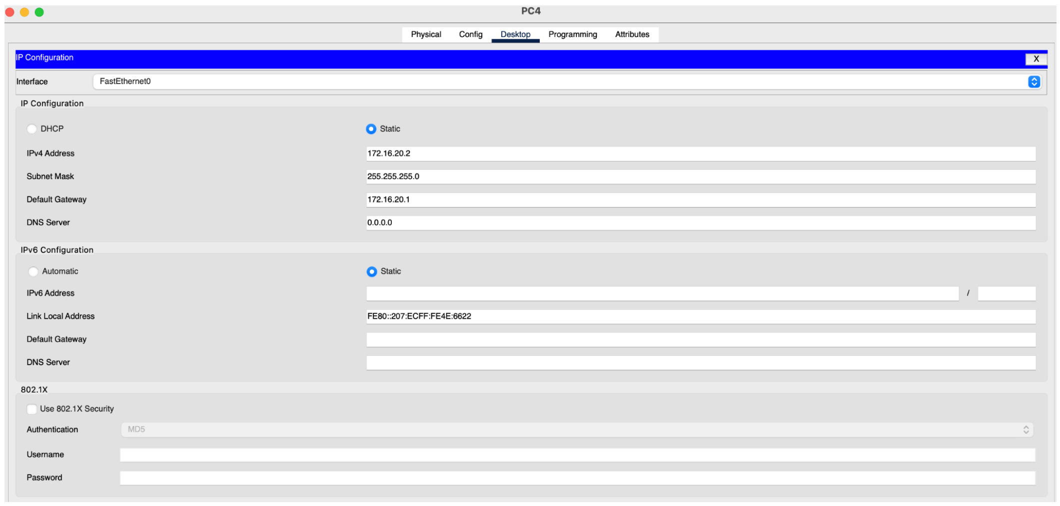
copy running-config startup-config

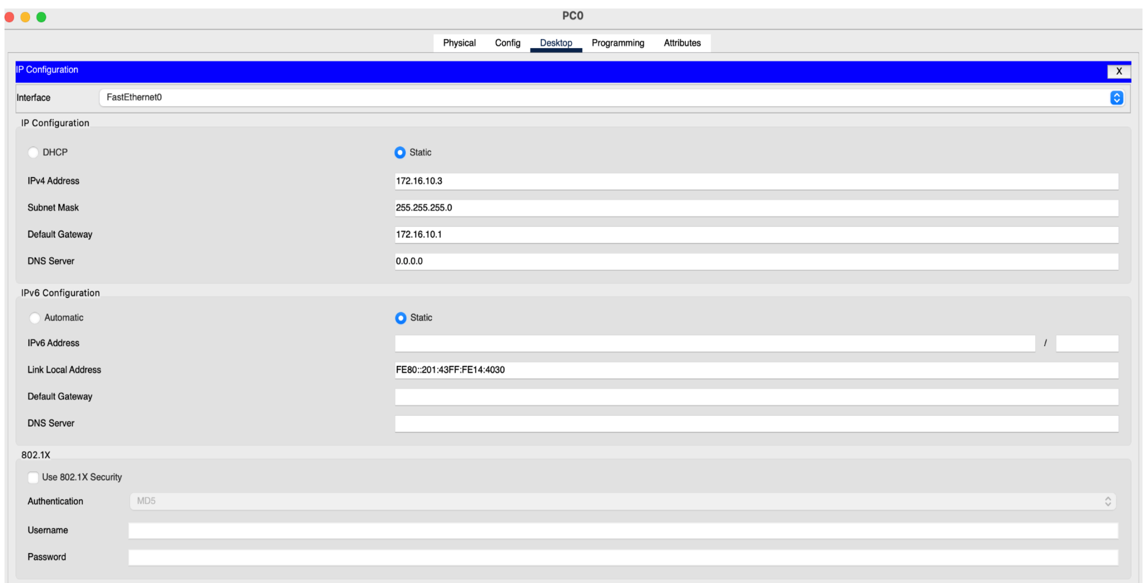


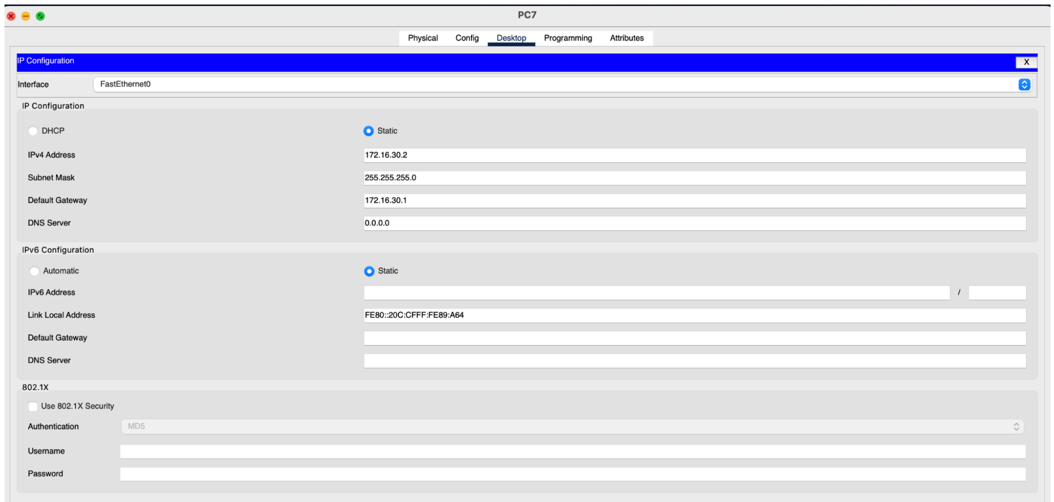

| Connection | Cable Type | Reason |
|---|---|---|
| PC to Switch | Copper Straight-Through | Used to connect different types of devices (PC and Switch). |
| Switch to Router | Copper Straight-Through | Used to connect different types of devices (Switch and Router). |

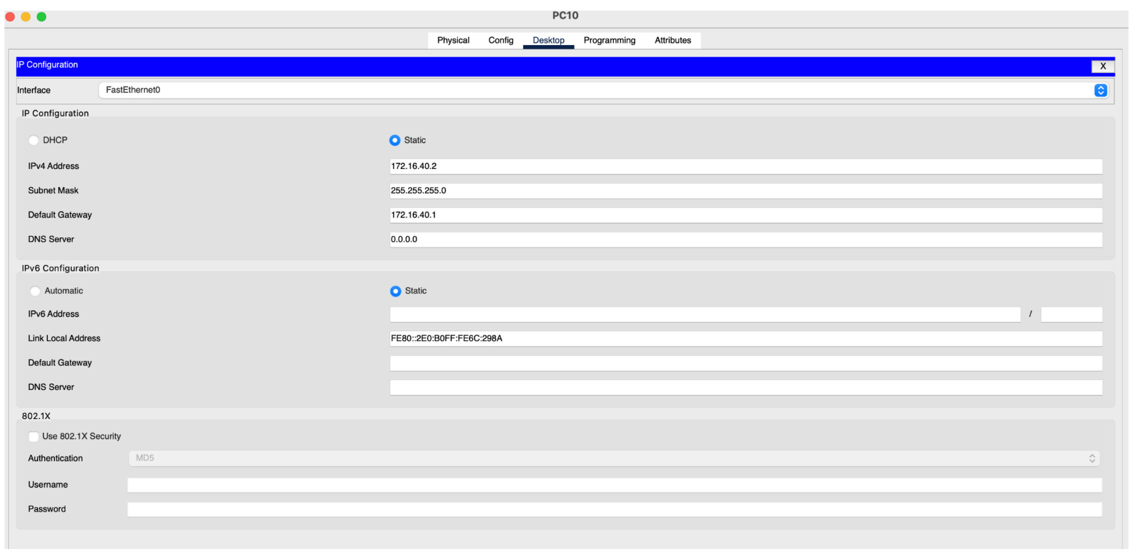
| LAN | Network | Subnet Mask | Default Gateway |
|---|---|---|---|

| | | | |
|---|---|---|---|
| ComputerLab | 172.16.10.0/24 | 255.255.255.0 | 172.16.10.1 |
| AdminOffice | 172.16.20.0/24 | 255.255.255.0 | 172.16.20.1 |
| Library | 172.16.30.0/24 | 255.255.255.0 | 172.16.30.1 |
| FacultyOffice | 172.16.40.0/24 | 255.255.255.0 | 172.16.40.1 |


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.20.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::207:ECFF:FE4E:6622
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.10.3
Subnet Mask
255.255.255.0
Default Gateway
172.16.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:43FF:FE14:4030
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.30.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.30.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20C:CFFF:FE89:A64
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC10
Physical
Config
Desktop
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Attributes
IP Configuration
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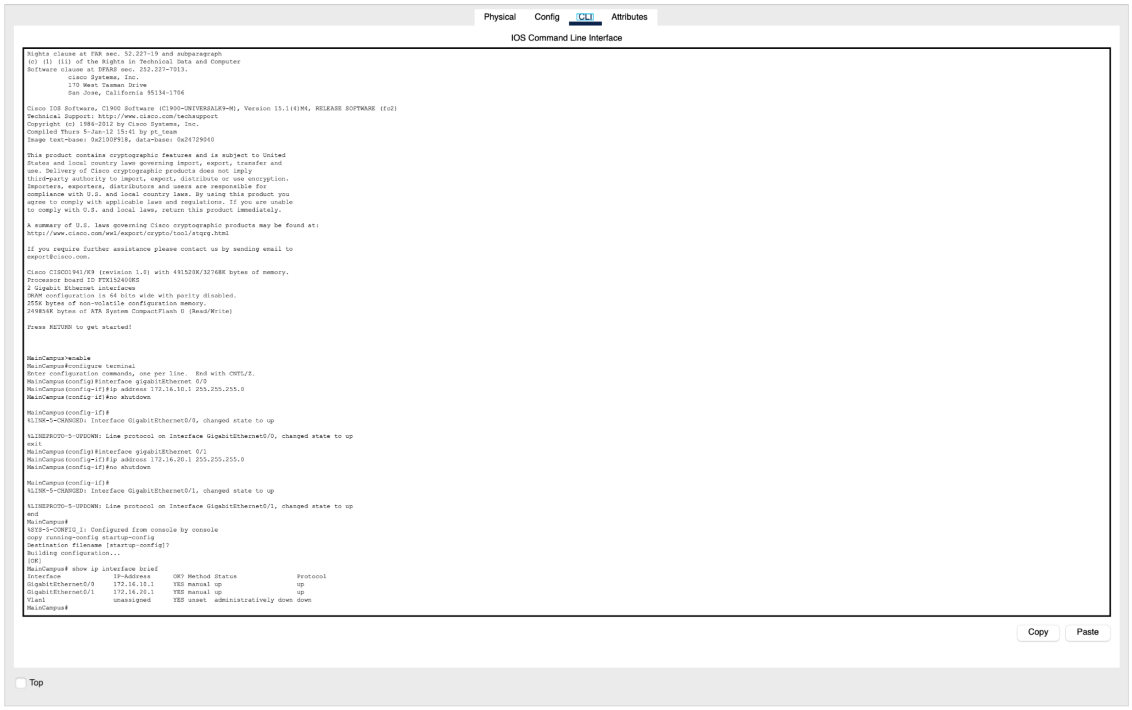
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.40.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.40.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2E0:B0FF:FE6C:298A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password




enable

configure terminal

interface gigabitEthernet 0/0

ip address 172.16.10.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 172.16.20.1 255.255.255.0

no shutdown

end

copy running-config startup-config

## Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق على النص الذي ترغب في أن يظهر هنا 1.


enable

configure terminal

interface gigabitEthernet 0/0

ip address 172.16.30.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 172.16.40.1 255.255.255.0

no shutdown

end

copy running-config startup-config

# Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق 1 على النص الذي ترغب في أن يظهر هنا.

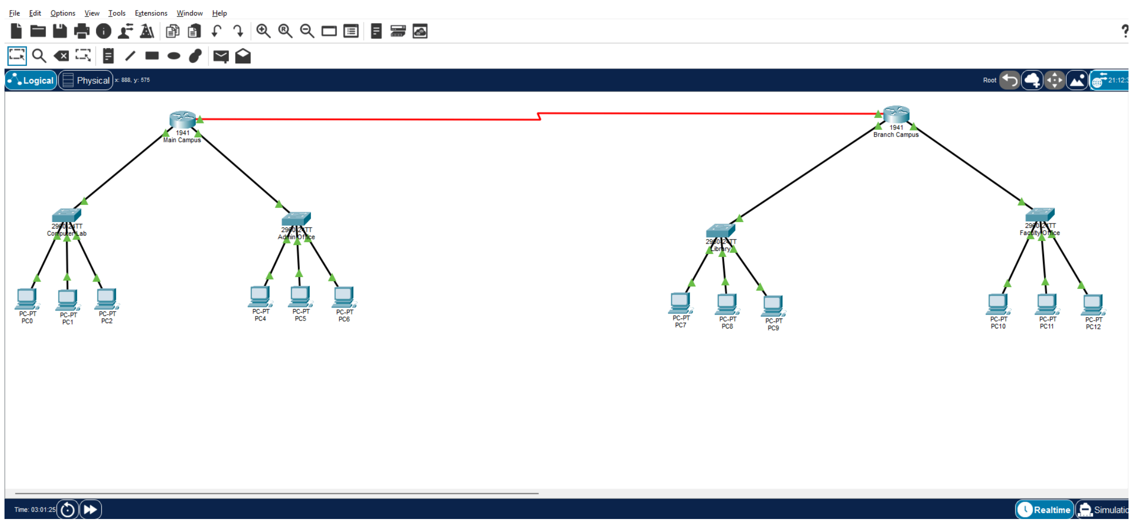



Figure - Final Design of the Network

**Commands for configuring the serial interface and enabling routing:**

Router 1 (**Main Campus**):

/Serial interface configuration

enable

configure terminal

interface serial0/1/0

ip address 10.10.10.1 255.255.255.252

clock rate 64000

no shutdown

exit

/Static routes

ip route 172.16.30.0 255.255.255.0 10.10.10.2

ip route 172.16.40.0 255.255.255.0 10.10.10.2

# خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق Heading 1 على النص الذي ترغب في أن يظهر هنا.


Router 2 (**Branch Campus**):

/Serial interface configuration

enable

configure terminal

interface serial0/1/0

ip address 10.10.10.2 255.255.255.252

no shutdown

exit

/Static routes

ip route 172.16.10.0 255.255.255.0 10.10.10.1

ip route 172.16.20.0 255.255.255.0 10.10.10.1

**Data Transmission using the Ping Command:**


Figure - From Main Campus to Branch Campus

ping 172.16.30.2


Figure - From Branch Campus to Main Campus

## ping **Version History / Commit Log**

The following internal record summarizes the main stages represented in the submitted report and Packet Tracer file.

| Version | Phase | Work Completed | Status |
|---|---|---|---|
| 1.0 | Phase 1 | Created the initial topology, selected devices, and named routers and switches. | Completed |
| 1.1 | Phase 1 | Connected LAN devices and prepared the IPv4 addressing plan for all four LANs. | Completed |
| 2.0 | Phase 2 | Configured LAN interfaces, serial addressing, clocking, and static routes. | Completed |
| 2.1 | Phase 2 | Verified bidirectional communication with successful ping tests and 0% packet loss. | Completed |
| 2.2 | Final | Reviewed commands, added reflection and artifacts, and prepared Word and PDF files. | Completed |

172.16.10.3

## Reflection

Finishing this project shifted how we think about troubleshooting, mostly because the whole layout only made sense if every piece behaved like part of a single system. The biggest hurdle showed up after we had all four LANs in place: getting consistent, dependable communication between the Main Campus and the Branch Campus. Inside each LAN, things worked without much effort. Crossing the WAN was different. Those first end-to-end tests forced us to slow down and verify the basics carefully, router interfaces, the serial connection, default gateways, and the routing table entries that would actually carry traffic from one side to the other.

To track it down, we went back to the addressing table, pulled interface status and configuration outputs from the routers, and traced the route one step at a time. Both ends of the serial link were brought up properly, the DCE side was set with a 64000 clock rate, and we double-checked every static route that needed to exist. After that, we reran the pings in both directions. Seeing four replies and 0% packet loss was the clear sign that the LAN settings and the WAN link were finally working together the way they were supposed to.

We chose static routing instead of RIP because the network is small, just two routers and four known LANs, and nothing about the design called for dynamic updates. With static routes, the forwarding path stayed obvious: each destination network and its next hop could be confirmed directly, without guessing what the protocol had learned. RIP might help if the topology expanded later since it cuts down on manual route edits, but here it would have meant extra configuration and constant routing updates that didn't add much value. For addressing, we kept the LANs on /24 subnets because they were simple to manage and easy to tell apart. On the serial link, we used a /30 since a point-to-point connection only needs two usable addresses. The tradeoff is straightforward: static routing doesn't scale well, and adding more campuses or networks would mean more manual changes.

Any AI help stayed limited to polishing the final text, checking formatting, and spotting minor presentation issues in the way commands were written. It wasn't used to run Packet Tracer or to do the actual configuration work for the group. Whenever a technical change came up during editing, we only kept it after matching it against the real topology, router output, IP settings, and, most importantly, the successful ping results.

If we were to do the project again, we'd test and document every stage immediately after finishing it instead of waiting until the end. That would mean cleaner screenshots for interface status, routing tables, and ping output, plus a more structured troubleshooting checklist. We'd also keep more frequent local copies of the report and the Packet Tracer file. With that approach, errors would be easier to isolate, and the final submission would end up stronger and more consistent.

## Project Artifacts

Cisco Packet Tracer project file (.pkt): IT351_FinalDesign.pkt (attached separately).

Final written report: IT351_Project-Phase_2_Final.docx.

Final PDF report: IT351_Project-Phase_2_Final.pdf.

Google Doc: Project-Phase2

GitHub/GitLab repository: GitHub

Packet Tracer project file (.pkt): Packet Tracer

## Final Submission

| Submission Item | Status |
|---|---|
| Final Word report (.docx) | Prepared |
| Final PDF report (.pdf) | Prepared |
| Cisco Packet Tracer project (.pkt) | Attached separately |